C1.2 Cell respiration

Notes written for this site. Slides, videos, worksheets and self-check questions: whatthecell.org/subtopics/sl-c1-2.html

Written for this site from the class slides. The chapter is on pages 405–411 of the course textbook. Image credits are on the last slide of the slides for this subtopic.

C1.2.1 ATP distributes energy within cells

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ATP (adenosine triphosphate) is a nucleotide. It is soluble, so it moves easily around a cell. It cannot pass through membranes, so it stays where it is needed. It also releases a useful quantity of energy, enough for many tasks.

C1.2.2 What ATP is used for

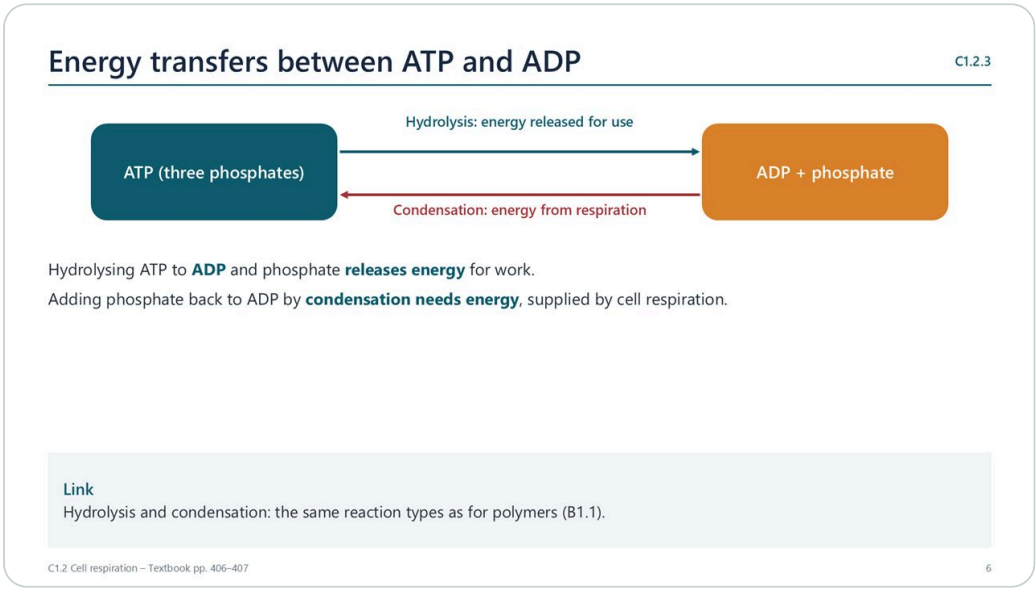
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- **Active transport** across membranes
- **Synthesis of macromolecules**, which is anabolism
- **Movement** of the whole cell, or of components such as chromosomes

C1.2.3 ATP and ADP

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Hydrolysis of ATP to ADP and phosphate releases energy. Energy from respiration is required to make ATP from ADP and phosphate.



C1.2.4 Respiration as a system for producing ATP

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Respiration uses the energy that carbon compounds release to produce ATP. Glucose and fatty acids are the main substrates, but many others can be used.

Watch out: respiration is a set of reactions in cells. Gas exchange is the movement of oxygen and carbon dioxide across a surface.

C1.2.5 Anaerobic and aerobic respiration in humans

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	ANAEROBIC	AEROBIC
WORD EQUATION	glucose → lactate	glucose + oxygen → carbon dioxide + water
SUBSTRATES	Glucose only	Glucose, fatty acids and others
YIELD OF ATP	Small, but fast	Large
WASTE PRODUCTS	Lactate	Carbon dioxide and water
WHERE	Cytoplasm	Cytoplasm and mitochondria

C1.2.6 Variables and rates of respiration

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Temperature, substrate and oxygen supply all affect the rate. A **respirometer** measures oxygen uptake: a CO_2 absorber removes carbon dioxide, so a drop of liquid moves as oxygen is used. If a drop moves 45 mm in 15 minutes in a tube of 1 mm^2, the rate is 3 mm^3 per minute.

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